

Model Validation Report Template

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1 Executive Summary

1-2 page summary

- Purpose of review
- Model description
- Current/intended model usage
- Validation methodology and scope
- Critical analysis

2 Introduction

- Pricer and/or functions reviewed (including version ID), including associated production system (e.g., Calypso)
- Business unit
- Current and intended model usage (e.g., marking and risk management of USD Bermudan swaptions)
- Report purpose
- Relevant history (models & validations) (e.g., update of model, last review was...)

3 Product Description

- Description of product
- Payoff function
- Example term sheets or live deals

4 Model Description

Reference earlier validation docs as appropriate

- Modeling theory/assumptions
 - Include critical discussion on model choice pros/cons as appropriate
 - Reference literature as appropriate
 - Describe any approximations made and assess their significance
- Mathematical description
 - Model inputs
 - Model outputs/reports (including later transformations if known/relevant, e.g., omega to vega transformation) and usage
- Model implementation, numerical methods (refer to quant docs as appropriate)
 - Critically discuss (e.g., impact of number of MC sims, integration errors, etc.)
- Calibration methodology (separate calibration tools require validation)
- Model usage (refer to desk docs if available)
- Model exposure (perspective on current and potential future model usage)

5 Validation Methodology and Scope

- What is the scope of the validation
- Describe how validation was performed
 - Description of benchmark MRM model
 - Tests performed (seek to be comprehensive as possible in terms of model inputs and instruments tested, keeping in mind the intended model usage)
 - Outputs reviewed (e.g., NPV, delta, vega)
 - Other data reviewed such as Totem, P&L explains

6 Validation Results

- Presentation and critical discussion of test results and other data reviewed
 - graphs are excellent devices here for displaying anomalous behavior

7 Conclusions and Recommendations

- Summary of validation results
- Recommendations

8 Bibliography



Model Validation Municipal Bonds

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Executive summary

This is a review of Bloomberg's risk model for municipal bonds, as included in Bloomberg's Enterprise Risk product. The risk model is used to calculate historical VaR, historical stressed VaR, expected shortfall, and sensitivities.

Risk calculations have two components. The first component is the pricing model used for the calculation. The second component is how the pricing model is used to calculate risks.

For municipal bonds, our pricing model is an OAS model using the LGM model calibrated to the swap market for the short rate component. Our historical VaR and expected shortfall models are based on applying historical perturbations of model inputs to the current market and repricing.

We find that the model has the following strengths:

- The pricing model is an industry standard model and accurately prices municipal bonds.
- Interest rate risk, credit risk and interest rate volatility risk are well accounted for.

The model's weakness is in the discrepancies between the computation of historical rates and their usage in the pricing model.

Overall, we find that this weakness is immaterial for risk analysis. We conclude that our approach is suitable for risk computations.

Introduction

This is a review of Bloomberg's risk calculation model for municipal bonds, as included in Bloomberg's Enterprise Risk product. We cover the computation of historical VaR, historical stressed VaR, expected shortfall, and sensitivities.

Risk calculations have two components. The first component is the pricing model used for the calculation. The second component is how the pricing model is used to calculate risks.

We will document both components and analyze the suitability of the model for risk analysis.

Product description

Municipal bonds are bonds issued by US municipalities. They can have fixed coupons, floating rate coupons or be zero coupon bonds. They can be callable, putable, and have sinking funds. We do not model municipal bonds that have coupons set by a remarking agent (such as CUSIP 756872DU8).

The risk factors that affect municipal bond prices are interest rates and municipal credit. Those that are callable or putable are also exposed to interest rate and credit volatility.

Pricing model description

The system prices municipal bonds by discounting at a credit spread (OAS) over a stochastic interest rate. The interest rate model is a normal, mean reverting one factor short rate model with time dependent volatility, calibrated to the swap market. This short rate model is known as the Linear Gaussian Markovian model (LGM), and is equivalent to the Hull-White model.

Model dynamics

The Hull-White model specifies the risk neutral dynamics for the short rate r when using the money market account B as the numéraire. Under this numéraire, the short rate is a mean reverting normal process:

$$dr_t = (\theta(t) - ar_t)dt + \sigma(t)dW_t \quad (0.1)$$

and the money market account is given by:

$$dB_t = r_t B_t dt \quad (0.2)$$

The parameter a is the mean reversion, and is taken to be 0.03. The parameter $\sigma(t)$ is chosen to calibrate to interest rate options. The θ parameter is chosen to match the currently observed discount factors [Stein, 2009, Piza, 2005].

The LGM model with parameters $\alpha(t)$ and $H(t)$ is given under the risk

neutral measure with respect to the numéraire N by:

$$dX_t = \alpha(t)dW_t \quad (0.3)$$

$$d\eta(t) = \int_0^t \alpha^2(s)ds \quad (0.4)$$

$$N_t = \frac{1}{D(t)} e^{H(t)X_t + \frac{1}{2}H^2(t)\eta(t)} \quad (0.5)$$

The function $D(t)$ is the current discount function, which is also today's price for receiving \$1 at time t . Under this numéraire, $P(t, T)$, the time t value of receiving \$1 at time T (A.K.A. the zero coupon bond process for a bond maturing at time T), is given by:

$$P(t, T) = E_t^L \left[\frac{1}{N_T} \right] N_t \quad (0.6)$$

$$= \frac{D(T)}{D(t)} E_t^L \left[e^{-H(T)X_T - \frac{1}{2}H^2(T)\eta(T)} \right] e^{H(t)X_t + \frac{1}{2}H^2(t)\eta(t)} \quad (0.7)$$

$$= \frac{D(T)}{D(t)} e^{(H(t)-H(T))X_T + \frac{1}{2}(H^2(t)-H^2(T))\eta(T)} \quad (0.8)$$

where E^L is expectation under the LGM risk neutral measure with respect to numéraire N .

The equivalence between the LGM formulation and the Hull-White formulation is given by setting the LGM parameters to satisfy:

$$a = -\frac{H''(t)}{H'(t)} \quad (0.9)$$

$$\sigma(t) = H'(t)\alpha(t) \quad (0.10)$$

$$\theta(t) = -\frac{H''(t)}{H'(t)} \frac{d \log D(t)}{dt} + \frac{d^2 \log D(t)}{dt^2} - (H'(t))^2 \eta(t) \quad (0.11)$$

For constant Hull-White parameters, this gives the LGM parameters:

$$H(t) = e^{-at} \quad (0.12)$$

$$\alpha(t) = \frac{\sigma}{a} e^{at} \quad (0.13)$$

$$\eta(t) = \frac{\sigma^2}{2a^3} (e^{2at} - 1) \quad (0.14)$$

Since a constant mean reversion of $a = 0.03$ is used, H has the form as given in equation 0.12. Defining N as in equation 0.5 automatically calibrates the model to the observed discount factors. The parameter $\alpha(t)$ is chosen to calibrate to a selection of observed option prices.

Model calibration

The pricing model inputs are:

- USD swap curve
- The credit spread (OAS) of the bond
- USD swap market volatility cube

The curve $\alpha(t)$ is nonparametrically chosen to calibrate to a strip of swaptions selected from the volatility curve. The swaptions chosen are the ones with exercise date matching the bond's call option exercise dates, with maturity matching the bond's maturity and coupon matching the bond's coupon, adjusted for off-par call strikes. If the bond is also amortizing or has a step coupon, the strike and tenor for each exercise date are chosen so as to match the duration and convexity of the remainder of the bond.

Once the interest rate model is calibrated, the OAS is computed to price the bond to the currently quoted mark to market.

Model implementation

The LGM model is implemented using a convolution to roll back prices from one cash flow date to the previous cash flow date. This requires using above formula for the forward prices of the zero coupon bonds.

Curve handling

The USD swap curve is stripped two different ways. For pricing, it is assumed that the discount function is given by

$$D(t) = e^{-tZ(t)} \quad (0.15)$$

with the spot rate curve $Z(t)$ being piecewise linear, continuous, and linear between consecutive maturities of instruments on the swap curve. The spot

rates are chosen so that swaps at the quoted rates are at the money when the forward LIBOR rates are projected on the curve.

Alternatively, the user can choose to discount based on the OIS curve instead of based on the USD swap curve, but not with dual curve stripping.

Historical forward rates are based on dual curve stripping, using the USD swap rates along with OIS rates. The OIS curve is used for discounting, and the forward LIBOR rates are computed so that swaps at the quoted rates are at the money when both legs are discounted according to the OIS curve.

The swap curve is expressed as the forward rates between selected maturities, which are 3 months, 6 months, 1 year, 2 year, 3 year, 4 year, 5 year, 7 year, 10 year, 20 year and 30 year. Let these maturity times (in years) be t_i , so $t_1 = 0.25$, $t_2 = 0.5$, $t_3 = 1.0$, etc. Let $t_0 = 0$. Let $Z(t)$ be the zero rate for time t (in years). Let f_i be the i^{th} forward rate. Then f_i is the forward rate from time t_{i-1} to time t_i (e.g. the $t_i - t_{i-1}$ tenor rate for time t_{i-1}) as a simple interest rate. This is the rate that satisfies:

$$1 + (t_j - t_{j-1})f_j = e^{t_{j-1}Z_j(t_{j-1}) - t_j Z_j(t_j)} \quad (0.16)$$

Risk model description

Risk calculations provided are value at risk (VaR), stressed value at risk, expected shortfall, and sensitivities to moves of the interest rate curve, the credit credit spread, and the volatilities.

VaR computations

The system computes historical VaR, stressed VaR, and expected shortfall. The client can choose the historical VaR window size, the stressed VaR window size, the horizon time, and the percentile level.

The VaR for is computed using a window of history that ends with the most recent quotes. For a 1-day horizon VaR, the daily changes are applied to the model inputs and the model is recalibrated and used to compute the perturbed price of the bond.

The changes applied are the changes in the USD swap curve, the USD swap market implied volatilities, and the change in average credit spread for municipal bonds of the same credit rating and maturity.

Let the business dates that lie in this window be d_i , $1 \leq i \leq n$, and let today be d_{n+1} . For a one year window, this would make n approximately 252.

For the swap curve, the relative change in the history between the forward rates that comprise the swap curve is applied to the forward rates comprising the current swap curve. Specifically, if the i^{th} forward rate for date j is $f_{i,j}$, then the rates used for the scenario corresponding to the changes from date d_j to date d_{j+1} are:

$$\bar{f}_{i,j} = \frac{f_{i,j+1}}{f_{i,j}} f_{i,n+1} \quad (0.17)$$

The volatilities are also perturbed by relative changes. For each maturity, implied volatilities are shifted for all coupons and all tenors. The shift applied is the the change in the implied volatility of the swaption to enter into a 5 year swap at the corresponding forward swap rate (an at the money swaption), exercisable at the same maturity.

The base OAS used for all calculations is the OAS that prices the bond to the client's mark to market for that bond.

The base OAS is perturbed by the relative change in credit spread for municipal bonds of the same maturity and rating. Let the current time to maturity of the bond be M years. The credit spread for an M year bond on each date d_i is estimated as the difference between the spot rate M years out at time d_i (computed using dual curve OIS stripping), and the spot rate M years out at time d_i derived from the municipal par yield curve computed on date d_i .

For a 1 year window and a 99th percentile VaR, given that n is approximately 252, the 99th percentile VaR would be given by the loss on the 2.52nd worst scenario. This is approximated by using the loss on the 3rd worst scenario, which corresponds to a 98.81st percentile VaR.

The expected shortfall is the average loss on all the scenarios that are worse than the scenario yielding the VaR. For a 1 year window and a 99th percentile VaR, this would then be the 2 worst scenarios.

The stressed 1-day VaR is the maximum of the 1-day historical VaRs for every historical VaR window whose dates lie within the stressed VaR window. For example, the stressed VaR window could be from January 1, 2007 to today. If using a 1 year historical VaR window, then stressed 1-day VaR would be the maximum of historical VaRs across all 1 year windows between January 1, 2007 and today.

The client can choose to calculate an n -day horizon VaR either based on historical n -day returns or based on a client specified scaling factor applied to the 1-day horizon VaR. For example, a scaling factor of $\sqrt{n}$ is typically chosen to scale a 1-day horizon VaR up to an n -day horizon VaR by virtue of the fact that this is how standard deviation scales for Brownian motion. The client can choose to be more conservative by using a larger scaling factor.

If computing n -day horizon VaR based on n -day returns, then the same is done for the stressed VaR and the expected shortfall. If computing n -day horizon VaR by a scaling factor, then the same scaling factor is used for stressed VaR and expected shortfall.

Sensitivities

The client can choose to compute interest sensitivity in one of two ways. The preferred method is to add one basis point to the swap curve and to recalibrate and recompute the price.

Alternatively, interest rate sensitivity can be computed on a yield basis. Given the price of the bond, the yield to worst is computed. The yield to worst is then shifted up by one basis point and the price is recomputed.

Credit sensitivity is computed by shifting the OAS up by one basis point and recomputing the price using the above model.

Volatility sensitivity (vega) is computed by adding 1% to all of the implied volatilities in the volatility cube, recalibrating the above model, and recomputing the bond's price.

Analysis

The modeling for computing risk on municipal bonds makes the following assumptions:

1. Interest rate modeling – Interest rates can be modeled by the LGM model calibrated to the USD swap curve and selected swap market implied volatilities.
2. Credit modeling – Municipal bond credit can be modeled as an additional spread over the swap curve.
3. Credit volatility modeling – The volatility of credit spreads need not be modeled in the pricer.

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4. Credit spread estimation – A good estimator for the municipal bond credit spread changes is the above noted algorithm based on the par curve for municipal bond yields and the swap curve stripped with OIS funding.
 5. Applying relative changes – Historical changes should be applied to current market risk factors on a relative basis.
 6. Default and migration modeling – Defaults and migrations do not need to be directly modeled in the pricer.
 7. Window size – The chosen window size is sufficiently large to yield an accurate historical VaR.
 8. IR sensitivity – The interest rate sensitivity method accurately estimates actual interest rate sensitivity.
 9. Dual curve stripping – Historical credit spreads changes and interest rate changes based on dual curve stripping can be applied to the current swap curve without dual curve stripping.

We will address each of the above assumptions.

Interest rate modeling

There are two basic approaches to option pricing. One is to develop a comprehensive model that captures all absolute and relative movements of all the risk factors. The other is to use a simple model that calibrates to the liquid options that most closely replicate the option being priced.

The first approach is appealing from a theoretical point of view, but presents a number of practical problems. Models that are complex enough to capture all absolute and relative movements require many more calibration instruments than are typically traded on the market, leading to the need to make a large number of modeling choices.

For example, for such a model to properly price interest rate spread options, one needs to know the risk neutral correlation between rates. Since such options rarely trade, this becomes an input that has to be estimated or set by a trader, rendering the pricing somewhat arbitrary.

Moreover, in a complicated model calibrated to the entire market, moves of each particular calibration instrument impact option prices, implying overly complicated hedges.

The second approach is less appealing from a theoretical point of view, but works well in practice. The theoretical objection is that it yields different dynamics for the same process when pricing different options. But in practice it works well in that it extrapolates in a realistic way from known option prices.

The most common example of the second approach is pricing equity options using Black-Scholes with a strike and maturity dependent implied volatility.

Here, the second approach is followed. Using the LGM model calibrated to the most appropriate hedging instruments is an interest rate modeling approach that is well accepted in industry. It is simple, straight forward, gives reasonable hedges, and does a good job of marking exotic options to market.

Normal interest rate models are sometimes criticized for producing negative rates. In practice, the appearance of negative rates is partially mitigated by the mean reversion, which bounds the long term short rate volatility. It is further mitigated by the calibration. By calibrating to the swap market options that most closely replicate the optionality structure of the municipal bond, it is guaranteed that the interest rate model is exactly calibrated to the most relevant subspace of the swap market, namely the options that would best hedge the interest rate volatility risk embedded in the bond.

The mean reversion parameter is not explicitly calibrated to the market. Instead, as has been a common street convention since the 1980s, a mean reversion of 3% is used. There is some justification in the literature for this value. In particular, in their figure 8, Gurrieri, Nakabayashi and Wong show historical mean reversion levels ranging from around 0% to 6%, with 3% being roughly a reasonable overall average [Gurrieri et al., 2010]. It is also well known that large changes in the mean reversion parameter have little impact in overall accuracy [Meng et al., 2013]. As a result, we believe that for short rate modeling, a mean reversion of 3% is reasonable to use with the Hull-White model.

Credit modeling

The credit risk is modeled as an OAS spread over the swap curve. This is a discount spread. While it is common to account for credit risk in this fashion, it is somewhat inaccurate because it is roughly equivalent to assuming recovery is on a percentage of the market price whereas in the bond market, recovery is on the par value.

Since the calibration is to par bonds, the model is accurate for bonds that are close to par. Pricing errors will grow as bonds deviate from par [Stein and Lee, 2011].

But these pricing errors will not show up directly in risk analysis. They are adjusted for in the OAS. This is completely accurate in that if the swap curve moves as per the scenario at hand, and the OAS changes as specified, then the bond price will be as specified.

The impact of this modeling choice is that the OAS includes a component dependent on how close the bond is to par. This means that using the par spreads to estimate the OAS will underestimate its volatility. But this will have a small impact on the actual pricing and risk calculations.

What is more important in risk analysis is to make sure that the model inputs are properly perturbed, which is the subject of subsequent subsections.

Credit volatility modeling

The pricing model does not model credit volatility. This is immaterial because it is reasonable to assume that default and interest rates are independent, in which case pricing is independent of credit volatility.

Credit spread estimation

One weakness of this approach is in the estimation of the credit spreads. Instead of using the entire credit spread curve (namely the municipal bond yield curve) as an input to the model, it extracts a spot rate spread from the curve for a particular maturity and uses that.

If using spot rate spreads, more appropriate would be to use it at the duration of the bond instead of at the maturity of the bond, but this introduced additional computation and complexity.

If the entire municipal bond curve cannot be taken into account, a better alternative would be to use the spread in par rates at the bond maturity

instead of the spread in zero rates, as it would be more representative of the corresponding bond behavior.

But, the impact of these approaches on estimated changes in credit spreads are small. Zero rates are relatively close to par rates, especially in flat interest rate environments. And changes in zero rates will match changes in par rates to first order. Hence the impact on VaR calculations is immaterial.

Note that using a credit rating based curve ignores idiosyncratic spread volatility. But this should not be taken into account in the market VaR, as it would be accounted for in the Specific Risk charge.

Applying relative changes

There is a flurry of literature on the subject of how to apply historical changes to current interest rates. Some argue that rates are log-normally distributed, so relative historical changes should be applied multiplicatively (as is done here). Others argue that rates are normally distributed, so absolute historical changes should be applied additively.

In fact, if the historical level of rates is around the current level of rates, then it is irrelevant whether relative changes are applied multiplicatively or absolute changes are applied additively. Both approaches will result in similar perturbations.

The real issue is how changes from low rate environments are applied to high rate environments, and visa-versa. When historical rates are low relative to current rates, absolute moves of just a few basis points will correspond to large relative moves. Applying the absolute moves to high rates will underestimate market moves and applying them multiplicatively will overestimate market moves.

Analogously, when historical rates are high relative to current rates, absolute changes of one or two percent would correspond to relatively small relative changes. Applying such absolute changes to a low interest rate environment would yield absurdly huge rate moves, and can also push rates deeply negative. Conversely, applying the relative changes would yield very little volatility.

Neither of the approaches is ideal, but applying relative changes is more robust and gives reasonable results.

Default and migration modeling

Default and migration are not modeled here, but they shouldn't be part of a market risk model. They are accounted for in the Incremental Risk Charge and by the Specific Risk charge.

Window size

The user can choose the window size for historical VaR calculations. A large enough window should be chosen to yield stable VaR calculations. When using a small window size, the VaR calculation will be accurate on average, but the daily changes in VaR will have a high variance. We would recommend a 3 year historical window for stability reasons, but a 1 year window could be used as well.

For stressed VaR, we would recommend a start date of January 1, 2007 or earlier, so that the stressed VaR window includes the last financial crisis.

IR sensitivity

Bumping the interest rate curve by a basis point and recomputing would be preferred for interest rate volatility because it would also account for the change in rates on the price of the embedded options. It also accounts for the shape of the yield curve.

Bumping yield to worst is analogous to computing sensitivity using a flat curve calibrated to the price and assuming zero volatility. This will be fairly accurate for noncallable and nonputtable bonds. It is less accurate for bonds with embedded optionality, but even so, we do not expect that it would have a material impact.

Dual curve stripping

Historical forward rates derived from dual curve stripping are applied to the forward rates derived by directly stripping the swap curve. This is an inconsistency. The correct approach would be to use dual curve stripping both for the historical forward rates as well as input into the model.

The impact of this inconsistency is immaterial. The relative changes of the dual curve stripped forward rates would be about the same as for the

directly stripped forward rates. The methodology used would in fact slightly overstate the absolute change in rates for each scenario.

Conclusions

We conclude that this approach is adequate for VaR calculations. Its biggest weakness is how historical changes are computed and applied to the current market. This could be improved by making the following changes:

1. Use dual curve OIS stripping in the interest rate model, so that historical forwards are computed consistently with how rates are modeled in the interest rate model.
2. Perturb OIS rates along with the swap rates.
3. Use the difference in the par rates instead of the difference in the spot rates as a proxy for the credit spread, or use the municipal bond yield curve as an input to the interest rate model.
4. Better adaptation of historical changes to the current market – adjusting for applying low rate environment changes to high rate environments and visa-versa.
5. Perturb volatilities by changes in smile and skew as well as changes in ATM volatilities, taking tenor into account as well.

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